

Abdulrahman Munir

ML Engineer — Low-Resource NLP & ASR

Profile

ML engineer focused on **low-resource NLP and automatic speech recognition (ASR)**, with full stack engineering ability to ship the systems I build. Native **Hausa** speaker—grew up in Hausa-speaking communities, childhood education in **Arabic**—combining machine learning, Python/React/Node engineering and data analytics with real linguistic fluency to build **speech-to-text and transcription systems for Hausa**, a language underserved by commercial ASR. Fine-tuning Whisper, wav2vec2/XLS-R and shipping production voice/AI products. Open to internships.

Core Strengths

Speech & Audio	Automatic Speech Recognition (ASR), fine-tuning Whisper and wav2vec2/XLS-R, forced alignment, VAD and audio denoising, audio dataset curation, transcription and annotation
NLP	Low-resource NLP, language modelling, tokenization, transliteration (Arabic script ajami & Latin boko for Hausa), corpus building
Linguistic Edge	Native speaker of Hausa, childhood education and fluency in Arabic, strong command of English; deep familiarity with Nigerian accents and dialectal variation across the Hausa-speaking region
Full Stack	React, Node.js, Kotlin/Jetpack Compose, Python (FastAPI/Flask), REST APIs, SQL, responsive UI, end-to-end app development
Data Analysis	pandas, NumPy, SQL, data cleaning, visualization (Matplotlib/Seaborn), exploratory analysis and reporting
ML Engineering	PyTorch, TensorFlow/Keras, Hugging Face Transformers & Datasets, scikit-learn, pandas/NumPy, experiment tracking
Dev & MLOps	Python, Git, Docker, AWS (S3/EC2), Linux, Jupyter, Label Studio (data annotation), ffmpeg/audio tooling

Technical Skills

Languages	Python, JavaScript/TypeScript, Kotlin, Bash, (learning) C++	Frameworks	PyTorch, TensorFlow, React, Node.js, Jetpack Compose
Speech/ASR	Whisper, wav2vec2/XLS-R, Librosa, Torchaudio	NLP	NLTK/Hugging Face NLP, tokenizers, low-resource pipelines
Data	pandas, NumPy, SQL, Datasets, Label Studio, visualization	Audio/Dev	ffmpeg, Sox, Docker, Git, AWS, Linux

Projects

Abuja – Nigeria

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GitHub: github.com/kingvirus0

2024 – **Hausa Speech-to-Text Transcription System**, *Independent project, Python, Whisper, Hugging Face, PyTorch*

- Built end-to-end **Hausa ASR pipeline** handling audio ingestion, VAD, denoising, and timestamped dual-script output (Latin *boko* and Arabic *ajami*).
- Fine-tuned **Whisper** and **wav2vec2-XLSR** on the low-resource **Mozilla Common Voice Hausa** corpus plus self-collected recordings.
- Curated a custom Hausa audio dataset with **forced alignment** and quality filtering, improving conversational speech transcription accuracy.
- Applied augmentation, orthographic normalisation and language-model rescoring to reduce out-of-vocabulary errors in a low-resource setting.

Jan 2024 **Hausa–Arabic Transliteration & Script Tool**, *Independent project, Python, rule-based + neural mapping*

- Built a tool converting between Hausa **boko** (Latin) and **ajami** (Arabic-script) writing systems, leveraging childhood Arabic literacy.
- Designed a phonetic mapping table plus a small alignment model to support transcription and linguistic search.

Experience

2025 **Image Annotation for Object Detection (Contract)**, *Independent / client-contracted, Nigeria*

- Labelled object bounding boxes in images using **LabelImg**, exported in Pascal VOC (XML) and YOLO formats for model training.
- Applied consistent label conventions; reviewed and corrected annotations to minimise mislabelled boxes and keep data quality high.
- Produced a clean, labelled dataset that fed directly into an object-detection model pipeline.

Education

2023 – **B.Eng. Computer Engineering**, *Federal University of Technology, Minna, Nigeria*

Present • Relevant coursework: Machine Learning, Deep Learning, NLP, Data Structures, Statistics.

Languages

English	Professional proficiency (working language)	<i>Read, write, speak</i>
Hausa	Native (fluent, incl. dialectal variants)	<i>Read, write, speak</i>
Arabic	Native-level from childhood education	<i>Read, write, speak</i>

Personal Projects

2024 – **Athena — Voice Assistant with Speech-to-Text**, *Personal (voice/AI assistant), Python, React, STT/TTS*

- Building a local-first AI voice assistant (Python backend, React UI) with on-device **speech-to-text** transcription, text-to-speech and a "*Hey Athena*" wake-word demo.

2024 – **Qur'an Recitation Video Generator**, *Personal, Python, ffmpeg, SRT timing*

Present • Python generator rendering recitation videos with **word-by-word Arabic subtitles synced to audio** (SRT), plus a timing editor aligning transcript segments to exact audio timestamps.

2024 – **ÌròyìnCheck — Multi-Platform News-Checking Bot**, *Personal, Node.js, Baileys, WhatsApp/Telegram/Discord*

- Fact-checking/news-checking bot delivered across **WhatsApp (Baileys v7)**, **Telegram** and **Discord** from a single resilient Node.js codebase.

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